

Assignment 1

What are main objective of information Security Audit?
Explain in your own word with one simple example.

An information security audit is done to check whether an organizations IT system are secure reliable & working properly to support business goal.

Main objectives

- Confidentiality - Ensure that sensitive data is accessible to only a authorised people
- Integrity - Verify that data is accurate, complete and not tampered.
- Availability - Check that system & data are available whenever needed for business operation
- Compliance - Confirm that IT practices follow law, policies and industry standard.
- Efficiency & Effectiveness - Assess whether IT resource are used optimally to meet business needs
- Risk Management - Identify & reduce risk such as data breaches, downtime or fraud

eg.

In a bank's online banking system, an IS audit will verify that only customers can access their accounts (confidentiality), transaction are recorded accurately without errors (integrity), & portal is always available for use without frequent downtime (availability).

Q.2 Write a short note on structure of IS audit report.
Why is it important for an organisation?

→ An IS audit report is formal document prepared after completing an audit of an organization's information system. It presents finding, risks and recommendations in structured way.

* Structure of IS Audit Report

- i) Title & Introduction - Mention the audit name, objectives, scope & period of audit
- ii) Executive Summary - A short overview of key findings, risks & recommendations (for management)
- iii) Audit Objectives & Scope - Explains what areas were audited (eg. n/w security, data handling, compliance)
- iv) Methodology - Describes how the audit was performed
- v) Finding & observations - Details strengths, weakness, & any security or control gap found.
- vi) Risk analysis - Explain potential impact of identified issues
- vii) Recommendations - Suggests corrective action to fix problem & improve security
- viii) Conclusion - Summarizes overall system health & compliance system status
- ix) Appendices - Supporting data, screenshots, log or ref.

* Importance of IS Audit Report

- Decision Making - helps to manage & understand IT risk & take informed action.
- Improves Security - Identifies vulnerability before attack exploit them
- Compliance - Proves adherence to legal requirement
- Risk management - Prioritizes risk & provides mitigation strategies.

What are computer Assisted Tools (CAAT). Give two examples & explain how they are useful.

- Computer Assisted Audit Tools (CAAT)
- CAAT are specialized software tools & techniques that auditors use to analyse large number of data stored in computer system.
- Instead of manually checking every record, CAATs help auditors test transaction, detect errors & identify fraud more quickly & accurately.

eg.

- 1) ACL (Audit Command Language)
 - A popular audit tool that allows auditors to import data from database, spreadsheet & ERP systems.
 - It can filter, sort & run tests on millions of records to detect irregularities like duplicate payment or unauthorised transactions.

- 2) IDEA (Interactive Data Extraction & Analysis)
 - data analysis software widely used in auditing.
 - It helps in extracting, sampling & analyzing data, making it easier to identify trends, fraud or issue.

How CAATs are used.

- Efficiency - Auditors can test large data quickly
- Accuracy - reduce human errors by automation
- Fraud detection - helps uncover hidden patterns or policy violation
- Compliance - Ensure that organization is following accounting standard for & regulatory requirement

Q.4 Explain difference between physical security control & logical security control with examples

Aspect protects	physical security hardware, buildings, personnel	logical security Data, software, networks
Nature	Tangible, real world measure	Intangible, software/system measure
Threat Handled	Theft, fire, natural disasters, unauthorised entry	Hacking, malware, data breaches, insider misuse
Access Control	Guards, ID cards, biometrics	Password, firewall, encryption
Example	CCTV, locks, fire alarms	Password, MFA, firewall

Q.5 Why user training & incident reporting important in org. security? Write any two points with examples.

→ 1) User training

- Why important - Even the best security tools fail if users don't know how to use them properly. Training employees helps them recognise threat like phishing emails, weak passwords or social engineering.

- example

- If employees are trained not to reuse weak password & to enable ~~multi~~ MFA, it reduces chance to hackers gaining unauthorised access to system.

- 2) Incident Reporting
- Even with strong security system, incident such as malware infections, suspicious logins or data leaks can still occur.
 - A well-established incident reporting process allows employees to quickly notify IT/security team, enabling faster detection & recovery.
 - Timely reporting prevents small issues from escalating into major breaches
 - eg. If employee reports that their system is behaving unusually (slow, strange pop-ups), the IT team can immediately investigate & isolate the affected computer, stopping potential threat from across org.

6. What are input, processing & output controls in application software? Explain with a small example.

→ 1) Input control

- Safeguards to ensure that the data entered into system is valid, accurate & complete before processing
- Prevent wrong or incomplete data from entering the system
- eg. data validation
Mandatory fields
Range check

2) Processing Controls

- Check applied while data is being processed to ensure accuracy, completeness & proper handling of transaction
- prevent error during completion, data manipulation, or transfer
- eg. System-generated logs to track each step of processing
- Automatic check sheet total balance (debits - credits)

3) Output Control

- Measure to ensure that result of processing are correct, complete & delivered to right person / system
- Prevent incorrect, incomplete or unauthorized output
- eg:
 - ATM prints receipt showing correct withdrawal amount
 - Confirmation emails of online shopping
 - Restricting who can view/download certain report

For 23/09/25